

Parvez Akhtar

+91 888-232-7732 | parvezakhtar218@gmail.com | [LinkedIn](#) | [GitHub](#) | [Kaggle](#)

PROFESSIONAL SUMMARY

Lead Engineer with 5+ years building scalable distributed systems, GenAI applications, and data platforms. Specializes in designing and deploying production-grade agentic AI workflows, RAG (Retrieval-Augmented Generation) systems, and LLM-powered microservices. Proven track record of reducing inference costs by ~90% through model fine-tuning and intelligent routing, while providing technical leadership and mentorship across cross-functional teams.

TECHNICAL SKILLS

GenAI & LLMs: Agentic Workflows & Multi-Agent Orchestration (LangChain, LangGraph, OpenAI Agents SDK, Anthropic SDK), RAG (Retrieval-Augmented Generation) with Hybrid Search & Re-ranking, MCP (Model Context Protocol), Tool/Function Calling, Structured Outputs, Prompt Engineering, Guardrails & Prompt-Injection Defense, Agent Memory & Human-in-the-Loop, LlamaIndex, Embeddings, Vector Databases (Qdrant, PGVector).

Model Training & Evaluation: Fine-tuning (LoRA/QLoRA, Hugging Face Transformers), Model Deployment & Serving, Evaluation Frameworks (RAGAS, Langfuse), PyTorch, TensorFlow, Scikit-learn, XGBoost.

Languages & Frameworks: Python, Java, TypeScript, SQL, FastAPI, Django, Spring Boot.

Cloud & Infrastructure: AWS (SageMaker, Bedrock, Lambda, ECS), Azure OpenAI, Docker, Kubernetes, Terraform, CI/CD Pipelines, Temporal.io.

Data Engineering: Apache Airflow, Kafka, Snowflake Cortex, Elasticsearch, ETL Pipelines, SQL, MongoDB, Redis, PostgreSQL, Browserbase.

Observability & LLMOps: LLM Tracing & Eval (Langfuse, OpenTelemetry, RAGAS), SigNoz, ELK Stack, CloudWatch, Prometheus.

PROFESSIONAL EXPERIENCE

Bigstep Technologies

Senior AI/ML Engineer, Lead Architect

Remote

04/2025 – Present

- As **Lead Architect**, drove GenAI platform delivery across client accounts spanning startups (Seed to Series F) and global enterprises including **CNBC London** and **VTS**, translating ambiguous business problems into scalable production architectures with technical mentorship of junior engineers.
- **GenAI Agent Design:** Architected a **Field-Aware Extraction Engine** processing **2M+ records/month**. Built a **Strategy Router** that classifies each field by complexity, routing ~60% to deterministic parsers and reserving the LLM for complex cases, cutting token spend ~40% with no accuracy loss.
- **RAG & Search Optimization:** Built a **Hybrid Search** pipeline fusing **SPLADE** (learned sparse) and **Sentence-BERT** (dense) retrieval via **Reciprocal Rank Fusion**, then layered **Cross-Encoder re-ranking**, improving **NDCG@5** by ~15 points (**0.71 to 0.86**).
- **Scale & Impact:** Scaled the GenAI platform from MVP to production, underpinning a **\$20M+ enterprise deal pipeline** within 6 months while optimizing model fine-tuning strategies for cost efficiency.
- **Agentic Workflows & Data Flows:** Orchestrated a serverless multi-agent framework for Salesforce automation with **tool/function calling** over **MCP (Model Context Protocol)** and swarm-based orchestration. Cut idle compute via a pure pay-per-execution model (saving ~\$45k/year).
- **Document Intelligence:** Built an agentic pipeline that abstracts structured terms from commercial real-estate leases with **structured-output validation**. Diagnosed a **10-point accuracy regression**, redesigning prompt engineering strategy and image preprocessing to recover the full drop and restore field-level extraction to ~88% baseline.
- **Data Engineering:** Constructed an entity matching pipeline on 22M records using an ensemble of **TF-IDF/Cosine Similarity** and 7 text metrics feeding an **XGBoost classifier** (85.3% AUC).

Kuttl (Incubated at IIM Lucknow)

Founding Engineer

Remote

12/2023 – 03/2025

- Joined as **Founding Engineer** for a GenAI-native orchestration platform; built the MVP from 0-to-1 that secured a term sheet. Documented and standardized the codebase and infrastructure, ensuring a **zero-downtime handover** to the incoming engineering team.
- **Model Fine-tuning & Deployment:** Fine-tuned **Llama-3-8B** using **LoRA** (Hugging Face **Transformers**) to replace GPT-4 for structured entity extraction, achieving **98% schema compliance** while reducing per-token inference costs by **90%** and latency by 300ms.
- **Scalable Data Workflows:** Engineered a distributed document parsing pipeline using **Python/FastAPI** and **Temporal.io**, processing 50k+ daily messages with resilient retry logic and dead-letter queues ensuring no data loss.
- **CI/CD & Model Evaluation:** Operationalized the ML lifecycle with **CI/CD evaluation gates** (RAGAS) scoring faithfulness and answer-relevancy, automatically blocking deployments that fell below threshold and catching quality regressions before production.

- **RAG Optimization:** Applied semantic caching and request batching for RAG endpoints, improving **P99 latency by 40%** during high-concurrency bursts.

Amplicomm

Bengaluru, India

SDE-2

05/2023 – 12/2023

- Led a team of 3 through a **Strangler-Fig migration** of a legacy monolith to **Spring Boot microservices**, enabling independent deploys with instant rollback. Enforced **code quality** gates via SonarQube that cut cyclomatic complexity ~42%.
- Established a centralized identity broker using **Keycloak** and Spring Security, replacing stateful DB lookups with **stateless JWT** validation to cut login latency by 30%.
- Engineered a Schema-per-Tenant architecture with RBAC for 21 enterprise clients, ensuring data integrity via automated checksum reconciliation pipelines.

Amazon

Bengaluru, India

Software Development Engineer (SDE)

03/2021 – 03/2023

- Owned the full SDLC for high-scale **microservices**, co-developing the SmartPOS billing engine and scaling it to support **3,000+ enterprise merchants** with 99.99% availability.
- Optimized **cloud infrastructure** costs by **25% (~\$100k/yr)** via strategic re-architecture of ECS services to utilize Spot Instances and custom auto-scaling policies on **AWS**.
- Developed a data-driven testing framework and **CloudWatch** service-level alarms, identifying **N+1 query bottlenecks** that cut p95 API latency by **20%** and surfaced performance regressions before release.
- Executed the zero-downtime migration of 6 core microservices from GCP to **AWS**, ensuring data consistency and seamless customer transition across **cloud platforms**.

Publicis Sapient

Remote, India

Associate Data Engineer

09/2020 – 03/2021

- Designed high-frequency **data ingestion pipelines** using **Apache Airflow** and Crawlera to scrape pricing data from Amazon, Walmart, and Target, bypassing complex anti-bot protections for competitive intelligence.
- Refined **Elasticsearch** indexing strategies and **SQL** aggregation queries, reducing query latency by ~40% to enable sub-second dashboard load times for brand performance analytics.

EDUCATION

National Institute of Technology Delhi

Delhi, India

B.Tech in Computer Science & Engineering

2016 – 2020

ACHIEVEMENTS

Amazon “On the Hero Road” Award: Recognized for exceptional impact on team delivery and engineering excellence.

Udacity Computer Vision Nanodegree: Full scholarship recipient (sponsored by Facebook).

Deep Learning Specialization: Completed 5-course specialization (Coursera) with perfect score.